

MATH 55 FIRST MIDTERM EXAM

Instructor: Nikhil Srivastava

February 19, 2026, 3:40-5:00pm, VLSB 2050

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. Do not unstaple the exam. Write your name and SID on every page. Show your work; numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.

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Sign here: _____

Question	Points
1	12
2	7
3	7
4	7
5	9
6	8
Total:	50

Do not turn over this page until your instructor tells you to do so.

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1. Select true or false (by filling in the circle) for each of the following. There is no need to provide an explanation.

- (a) (3 points) The compound proposition $(p \wedge \neg p) \rightarrow q$ is a tautology.

Answer: True

(The hypothesis $p \wedge \neg p$ is a contradiction, meaning it is always false. A conditional statement with a false hypothesis is vacuously true).

☐ True ☐ False

- (b) (3 points) The proposition

$$[\exists x \forall y P(x, y)] \rightarrow [\forall x \exists y P(x, y)]$$

is a tautology.

Answer: False

(Let $P(x, y)$ mean $x = 0$ in the integers. The premise $\exists x \forall y (x = 0)$ is true because 0 equals 0 for any y . But the conclusion $\forall x \exists y (x = 0)$ claims that for *any* x , there is a y such that $x = 0$, which is false for $x = 1$).

☐ True ☐ False

- (c) (3 points) If A, B, C are sets such that $A \subseteq B$, then $(A \times C) \subseteq (B \times C)$.

Answer: True

(If $(x, y) \in A \times C$, then $x \in A$ and $y \in C$. Since $A \subseteq B$, we have $x \in B$. Thus, $(x, y) \in B \times C$).

☐ True ☐ False

- (d) (3 points) If A, B, C are sets, then $A - (B \cap C) = (A - B) \cup (A - C)$.

(Here $A - B = \{x \in A : x \notin B\}$.)

Answer: True

(By De Morgan's laws for sets).

☐ True ☐ False

[Scratch Space Below]

Name and SID: _____

2. (7 points)

- (a) (4 points) Prove that if A and B are countably infinite sets, then $A \cup B$ must be countable.

Proof: Since A and B are countably infinite, their elements can be listed in sequences: $A = \{a_1, a_2, a_3, \dots\}$ and $B = \{b_1, b_2, b_3, \dots\}$. We can form a new sequence for $A \cup B$ by alternating elements from both sets: $s = a_1, b_1, a_2, b_2, a_3, b_3, \dots$. If there are any duplicate elements (since A and B might not be disjoint), we simply skip them when they appear again — to do this, let $s' = s'_1, s'_2, \dots$ be the sequence where s'_n is the n th new element of s (where s_i is new if $s_k \neq s_i$ for all $k < i$). The sequence s' contains all of the elements of $A \cup B$ and every element appears once by construction, so $A \cup B$ is countable. $\square$

- (b) (3 points) Find a counterexample to the following statement: if A and B are uncountable sets, then $A \cap B$ must be uncountable.

Counterexample: Let $A = (-\infty, 0]$ and $B = [0, \infty)$ be subsets of the real numbers. Both A and B contain $(0, 1)$, so they are uncountably infinite. However, their intersection is $A \cap B = \{0\}$, which contains exactly one element. Since $\{0\}$ is finite, it is countable.

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3. (7 points) Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.

(a) (4 points) Prove that if the composition $g \circ f$ is onto (i.e., surjective), then g must be onto.

Proof: To show that $g : B \rightarrow C$ is onto, we must show that for any element $c \in C$, there exists some $b \in B$ such that $g(b) = c$. Let c be an arbitrary element in C . Because the composition $g \circ f : A \rightarrow C$ is onto, there exists some element $a \in A$ such that $(g \circ f)(a) = c$, which by definition means $g(f(a)) = c$. Now, let $b = f(a)$. Since f maps A to B , $b \in B$. Substituting b into the equation yields $g(b) = c$. Therefore, for every $c \in C$, we have found a $b \in B$ such that $g(b) = c$, proving g is onto. $\square$

(b) (3 points) Find a counterexample to the following statement: if the composition $g \circ f$ is onto, then f must be onto.

Counterexample: Let $A = \{1\}$, $B = \{1, 2\}$, and $C = \{1\}$.

Define $f : A \rightarrow B$ by $f(1) = 1$.

Define $g : B \rightarrow C$ by $g(1) = 1$ and $g(2) = 1$.

The composition $g \circ f : A \rightarrow C$ maps $1 \mapsto 1$. Since 1 is the only element in C and $(g \circ f)(1) = 1$, the composition $g \circ f$ is onto. However, f is not onto because there is no element in A that maps to $2 \in B$.

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4. (7 points) Prove that the equation

$$n^2 + 1 = 3k$$

has no solution for integers n and k .

(Hint: Consider the corresponding congruence modulo 3).

Proof: Assume for contradiction that there are integers n, k such that $n^2 + 1 = 3k$. Reducing modulo 3, this implies that:

$$n^2 + 1 \equiv 3k \equiv 0 \pmod{3}$$

which upon rearrangement yields

$$n^2 \equiv -1 \equiv 2 \pmod{3}.$$

Let us consider all possible cases for the remainder of n modulo 3:

- **Case 1:** If $n \equiv 0 \pmod{3}$, then $n^2 \equiv 0^2 \equiv 0 \pmod{3}$.
- **Case 2:** If $n \equiv 1 \pmod{3}$, then $n^2 \equiv 1^2 \equiv 1 \pmod{3}$.
- **Case 3:** If $n \equiv 2 \pmod{3}$, then $n^2 \equiv 2^2 \equiv 4 \equiv 1 \pmod{3}$.

In every possible case, $n^2 \not\equiv 2 \pmod{3}$. Therefore, the congruence $n^2 \equiv 2 \pmod{3}$ has no integer solutions. Consequently, there are no integers n and k that satisfy $n^2 + 1 = 3k$. $\square$

5. (9 points) Let p be a prime number.

(a) (2 points) Prove that if n is an integer such that $p|n$, then $p|n^2$.

Proof: Let p be prime and let n be an arbitrary integer. By the definition of divisibility, since $p | n$, we can choose an k such that $n = pk$. Squaring both sides of this equation yields:

$$n^2 = (pk)^2 = p^2k^2 = p(pk^2)$$

Since k is an integer, pk^2 is also an integer, and we have written $n^2 = pm$ for an integer m , which by definition implies that $p | n^2$, as desired. $\square$

(b) (3 points) Prove that if n is an integer such that $p|n^2$, then $p|n$.

Proof: Let p be prime and let n be an arbitrary integer. By Euclid's Lemma, if a prime number p divides a product of two integers ab , then p must divide a or p must divide b (or both). Here, we are given that $p | n^2$, which can be written as $p | (n \cdot n)$. Applying Euclid's Lemma where $a = n$ and $b = n$, it must be true that $p | n$ or $p | n$. In either case, we conclude that $p | n$. $\square$

(c) (4 points) Prove that $\sqrt{p}$ is irrational. You may use the results from parts (a) and (b).

Proof: We will prove this by contradiction. Assume that $\sqrt{p}$ is rational. By definition, it can be expressed as a fraction of two integers in its simplest form:

$$\sqrt{p} = \frac{a}{b}$$

where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a, b) = 1$.

Squaring both sides gives:

$$p = \frac{a^2}{b^2} \implies a^2 = pb^2$$

Because b^2 is an integer, this equation shows that $p | a^2$. From the result proven in **part (b)**, since $p | a^2$, it must be true that $p | a$.

Because $p | a$, we can choose an integer k such that $a = pk$. Substitute $a = pk$ back into our equation $a^2 = pb^2$:

$$(pk)^2 = pb^2 \implies p^2k^2 = pb^2$$

Dividing both sides by p (since a prime $p \neq 0$), we get:

$$pk^2 = b^2$$

Because k^2 is an integer, this shows that $p | b^2$. Again, applying the result from **part (b)**, if $p | b^2$, then $p | b$.

We have now established that $p | a$ and $p | b$. This means p is a common factor of both a and b . However, this contradicts our initial assumption that $\gcd(a, b) = 1$, since $p \geq 2$. $\square$

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[Scratch Paper 1]

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6. (8 points)

(a) (3 points) Use the Euclidean algorithm to find $d = \gcd(123, 45)$.

Solution: We apply the division algorithm repeatedly:

$$123 = 45(2) + 33$$

$$45 = 33(1) + 12$$

$$33 = 12(2) + 9$$

$$12 = 9(1) + 3$$

$$9 = 3(3) + 0$$

The last non-zero remainder is 3. Therefore, $d = \gcd(123, 45) = 3$.

(b) (2 points) Find integers s and t such that $123s + 45t = d$.

Solution: We work backward through the steps of the Euclidean algorithm from part (a) (Bezout's Identity):

$$\begin{aligned} 3 &= 12 - 9(1) \\ &= 12 - (33 - 12(2)) = 12(3) - 33(1) \\ &= (45 - 33(1))(3) - 33(1) = 45(3) - 33(4) \\ &= 45(3) - (123 - 45(2))(4) = 45(3) - 123(4) + 45(8) \\ &= 45(11) - 123(4) \end{aligned}$$

So we have $123(-4) + 45(11) = 3$. Thus, the integers are $s = -4$ and $t = 11$.

(c) (3 points) Using your result from part (b), find an integer x such that:

$$45x \equiv 6 \pmod{123}$$

Solution: This is tricky, because 45 does not have an inverse modulo 123 since $\gcd(45, 123) > 1$. However, we can still solve the equation using modular arithmetic. From part (b), we know that:

$$123(-4) + 45(11) = 3$$

We want to solve $45x \equiv 6 \pmod{123}$. We can multiply our entire Bezout identity by 2 to get a 6 on the right side:

$$2 \cdot (123(-4) + 45(11)) = 2 \cdot 3$$

$$123(-8) + 45(22) = 6$$

Now, taking this equation modulo 123, the term $123(-8)$ evaluates to $0 \pmod{123}$, leaving us with:

$$45(22) \equiv 6 \pmod{123}$$

Therefore, a valid integer solution is $x = 22$.

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[Scratch Paper 2]

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[Scratch Paper 3]